

APM 1110 - FA 4

Cainglet

<https://github.com/Endermite50/APM1110-CAINGLET-FA-SA>

Item 2: Binary Communication Channel

A binary communication channel carries data as one of two sets of signals denoted by 0 and 1. Owing to noise, a transmitted 0 is sometimes received as a 1, and a transmitted 1 is sometimes received as a 0. For a given channel, it can be assumed that a transmitted 0 is correctly received with probability 0.95, and a transmitted 1 is correctly received with probability 0.75. Also, 70% of all messages are transmitted as a 0. If a signal is sent, determine the probability that: (a) a 1 was received; (b) a 1 was transmitted given that a 1 was received.

Given Probabilities

```
P_transmit0 <- 0.70
P_transmit1 <- 0.30

P_rec0_sent_0 <- 0.95          # P(receive 0 | sent 0)
P_rec1_sent_0 <- 1 - P_rec0_sent_0 # P(receive 1 | sent 0)

P_rec1_sent_1 <- 0.75          # P(receive 1 | sent 1)
P_rec0_sent_1 <- 1 - P_rec1_sent_1 # P(receive 0 | sent 1)
```

(a) Probability that a 1 was received

Using the law of total probability:

```
P_receive1 <- (P_transmit0 * P_rec1_sent_0) +
              (P_transmit1 * P_rec1_sent_1)
cat("P(1 received) =", P_receive1, "\n")
```

```
## P(1 received) = 0.26
```

(b) Probability that a 1 was transmitted given a 1 was received

Using Bayes' theorem:

```
P_transmit1_given_rec1 <- (P_transmit1 * P_rec1_sent_1) / P_receive1
cat("P(1 transmitted | 1 received) =", round(P_transmit1_given_rec1, 4), "\n")
```

```
## P(1 transmitted | 1 received) = 0.8654
```

Item 7: IT Company Error Rates

There are three employees working at an IT company: Jane, Amy, and Ava, doing 10%, 30%, and 60% of the programming, respectively. 8% of Jane's work, 5% of Amy's work, and just 1% of Ava's work is in error. What is the overall percentage of error? If a program is found with an error, who is the most likely person to have written it?

Given Probabilities

```
P_jane <- 0.10
P_amy  <- 0.30
P_ava  <- 0.60

P_err_jane <- 0.08 # P(error | Jane)
P_err_amy  <- 0.05 # P(error | Amy)
P_err_ava  <- 0.01 # P(error | Ava)
```

Overall probability of error

Using the law of total probability:

```
P_error <- (P_jane * P_err_jane) + (P_amy * P_err_amy) + (P_ava * P_err_ava)
cat("P(Error) =", P_error, "\n")
```

```
## P(Error) = 0.029
```

Probability each person wrote an erroneous program

Bayes's Theorem:

```
P_jane_given_err <- (P_jane * P_err_jane) / P_error
P_amy_given_err  <- (P_amy * P_err_amy)  / P_error
P_ava_given_err  <- (P_ava * P_err_ava)  / P_error

cat("P(Jane | Error) =", round(P_jane_given_err, 4), "\n")
```

```
## P(Jane | Error) = 0.2759
```

```
cat("P(Amy | Error) =", round(P_amy_given_err, 4), "\n")
```

```
## P(Amy | Error) = 0.5172
```

```
cat("P(Ava | Error) =", round(P_ava_given_err, 4), "\n")
```

```
## P(Ava | Error) = 0.2069
```

Most likely author of the error

```
probs <- c(Jane = P_jane_given_err, Amy = P_amy_given_err, Ava = P_ava_given_err)
most_likely <- names(which.max(probs))
cat("Most likely author:", most_likely, "\n")
```

```
## Most likely author: Amy
```